

Research Vision

My research vision is to make cellular networks predictable, measurable, and controllable to enable reliable next-generation latency-sensitive applications, including AR/VR, cloud services, and connected autonomy. I combine large-scale real-world measurements, cross-layer performance analysis, and programmable 5G experimentation to understand and address challenges in network reliability, latency, and performance. My goal is to develop lightweight, standards-compliant, and practical solutions that leverage AI-driven analytics and closed-loop network control to improve performance, energy efficiency, and reliability in real-world deployments.

Education

2021–2025 : **Ph.D. in Computer Engineering**, *Northeastern University*, Boston, USA.

2018–2020 : **Masters of Science in Computer Engineering**, *Southern Illinois University Carbondale*, IL.

Experience

2026 – 2026 **Research Engineer**, *AT&T Labs. Inc.*, contract.

- Developed a production-ready anomaly detection pipeline for E911 network KPIs using a lightweight Isolation Forest model, enriched with SIP/CCDR protocol context, including message types, cause codes, and failure indicators. Engineered historical baseline features using z-scores, rolling means, and rolling standard deviations to identify KPI degradation, achieving 90% anomaly correlation with known network incidents.
- Benchmarked graph-based SSB optimization approaches using Google OR-Tools and SCIP on production network inventory data, incorporating neighbor weights to evaluate optimization performance in terms of execution time, number of configuration changes, and iterations required to reach optimality.

2021 – 2023 **Research Intern and External Collaborator**, *AT&T Labs. Inc.*

- Investigated the impact on QoE of low-latency video streaming over 5G networks in a joint collaboration work with Facebook.
- Published a peer-reviewed conference paper correlating radio link dynamics with application-layer KPIs to proactively mitigate User Quality of Experience (QoE) degradation.

Projects

X5G: O-RAN compliant 5G Testbed Development, *NSF Funded*.

OAI
C/C++
O-RAN
NVIDIA ARC

- Contributed to the development of X5G: an O-RAN compliant multi-vendor (NVIDIA, OAI, Foxconn) programmable testbed for research.
- Instrumented OAI code for performance tuning, configuration tweaking, and KPI extraction.
- Developed measurement report and event (A3) handling mechanism for Xn-handover.
- Improved system throughput by 20% by debugging and bug reporting to Nvidia Aerial and OAI team.
- Published 1 Conference and 1 Journal Paper.

5G Measurement Studies on User-perceived Performance, *NSF Funded*.

5G/LTE
Python
X-CAL
Puffer
FFMPEG

- Designed and executed large-scale mobility measurement campaigns (multi-city/cross-country drives), integrating synchronized application-level tests with cellular KPIs for holistic analysis.
- Assessed 5G coverage, throughput, latency, and user Quality of Experience (QoE) using large-scale dataset [10TB+].
- Built an end-to-end measurement and QoE extraction pipeline for modern interactive applications, including AR/VR, 360° video streaming, and cloud gaming.
- Developed tools to extract lower-layer (PHY/MAC/RLC) KPIs from 5G control-plane signaling (RRC) [XCAL logs].
- Investigated key 5G mechanisms—handovers, uplink power control, and advanced features (carrier aggregation, beamforming, MIMO) to identify performance bottlenecks and explain observed behavior.
- Built automated Python pipelines for data cleaning, feature extraction, and statistical analysis; produced publication-quality figures and reproducible artifacts.
- Open-sourced **datasets/scripts** and created replication-friendly workflows to enable follow-on studies.
- Published 10 conference papers.

Throughput Prediction in Operational 5G Networks, CISCO Funded.

- 5G/LTE
Python
Machine
Learning
- Collected a fine-grained, real-world **dataset** during a 4-day (1000+ km) drive, measuring bidirectional application throughput at 100 ms time-scales across all three major U.S. operators and mixed LTE/5G deployments.
 - Evaluated multiple predictors (MLP, GDBT, transformer-based MVTs) alongside statistical baselines (EWMA, last-value) and showed that low RMSE can still yield poor application-centric accuracy (e.g., high ARE95 / low PARE10).
 - Used over-parameterized neural networks as a “learnability test” and found slow training convergence and persistently high validation/test error, indicating fundamental limits to fine-grained throughput predictability.
 - Identified conflicting sample sequences, where nearly identical historical throughput patterns were followed by drastically different next-sample values, forcing models to learn highly irregular mappings and limiting their generalization performance.
 - Published 1 Conference paper.

5G Performance Evolution Study, NSF Funded.

- JAVA
Android
Python
AWS
Machine
Learning
- Built an IRB-approved **Android measurement app** (NextG-UP) to collect uplink TCP throughput and RTT alongside cellular KPIs (GPS, RSRP/RSRQ/RSSI, Cell-ID, ARFCN/EARFCN, operator); implemented nuttcp (10s UL test) + ICMP ping (11 packets @200ms) with a lightweight footprint (6.5MB APK, <250MB RAM).
 - Deployed a geographically distributed measurement backend (AWS servers in N. Virginia, Oregon, Frankfurt + AWS Wavelength inside Verizon’s network in Boston) and integrated nearest-server selection + mobility modes (static/walking/driving).
 - Led a cross-sectional, year-long 5G uplink campaign with weekly volunteer measurements: 10,166 tests across 8 cities in 7 countries (2 continents), covering 12 operators and 1,422 unique cell IDs.
 - Quantified “deployment maturity” and found stagnation over one year (no clear throughput/latency trend) and inconsistent 5G advantage over LTE (e.g., 5G throughput lower in 1 city; 5G latency lower than LTE in only 3/8 cities).
 - Conducted the first “metamorphosis” lifecycle study of 5G—from initial rollout to maturity—tracking coverage, throughput, and latency while comparing against mature LTE and accounting for factors like device adoption and network load.
 - Identified 4-phase 5G coverage evolution (slow → rapid → second slow → stable) and 4-phase downlink throughput evolution, showing that coverage and throughput phases do not necessarily align.
 - Published 2 Conference paper and 1 Journal Paper.

Multipath TCP (MPTCP) for Dual-band 802.11ad + 802.11 ac networks, NSF Funded.

- C
Python
TCP
MPTCP
Android
Kernel
tc
802.11
- Ported MPTCP source code to a commercial dual-band smartphone (802.11ad@60 GHz + 802.11ac@5 GHz) by integrating 15,000+ LOC into the Android Linux kernel; released the implementation publicly for reproducible experimentation.
 - Built an experimental pipeline to measure **100 ms**-granularity throughput and on-device power draw, enabling a first look at both *performance* and *energy* of MPTCP on resource-constrained phones.
 - Identified and implemented key system-level optimizations for Gbps MPTCP on smartphones (e.g., enabling GRO and CPU/SIRQ affinity to big cores), achieving near sum-rate performance across the two radios under ideal conditions.
 - Quantified throughput–energy tradeoffs across packet-scheduling ratios and derived practical guidelines toward an energy-aware MPTCP scheduler for dual-band mmWave/legacy Wi-Fi phones.
 - Released two custom kernels with MPTCP stack for **ASUS ROG II** and **Pixel 5**.

Publications

In Conference Proceedings

- 2026 A Cross-Country Measurement Study of Inter-cell Interference in Cellular Networks PDF, Zhekun Yu, Yufei Feng, Moinak Ghoshal, Omar Basit, **Imran Khan**, Z. Jonny Kong, Sizhe Wang, Phuc Dinh, Y. Charlie Hu, and Dimitrios Koutsonikolas, *ACM CoNEXT*, 2026.
- 2026 Exploring the 5G Digital Divide in the Non-Contiguous US: LEOSatellites to the Rescue?, Sizhe Wang, Moinak Ghoshal, Yufei Feng, **Imran Khan**, Phuc Dinh, Omar Basit, Zhekun Yu, Y. Charlie Hu, and Dimitrios Koutsonikolas, *Proc. of ACM SIGMETRICS*, 2026.
- 2026 Uplink Power control in Operational 5G Networks: Configurations, Policies, Evolution, and Impact on Performance, **Imran Khan**, Yufei Feng, Moinak Ghoshal, Omar Basit, Z. Jonny Kong, Sizhe Wang, Phuc Dinh, Y. Charlie Hu, and Dimitrios Koutsonikolas, *Under submission*, 2026.

- 2026 Too Many Reports, Too Few Handovers: A Large-Scale Study of Configuration Inefficiencies in Operational 5G Networks,
Moinak Ghoshal, **Imran Khan**, Phuc Dinh, Z. Jonny Kong, Omar Basit, Sizhe Wang, Yufei Feng, Y. Charlie Hu, and Dimitrios Koutsonikolas, *ACM CoNEXT*, 2026.
- 2025 5G Metamorphosis: A Longitudinal Study of 5G Performance from the Beginning,
Imran Khan*, Omar Basit*, Moinak Ghoshal, Y. C. Hu, and Dimitrios Koutsonikolas, *Proc. of ACM IMC*, 2025.
- 2025 A First Large-Scale Study of Operational 5G Standalone Networks.,
Moinak Ghoshal, Omar Basit, Sizhe Wang, Phuc Dinh, **Imran Khan**, Yufei Feng, Zhekun Yu, Y. Charlie Hu, and Dimitrios Koutsonikolas, *Proc. of ACM CoNEXT*, 2025.
- 2025 Replication: Performance of cellular networks on the wheels,
Moinak Ghoshal, Omar Basit, **Imran Khan**, Z. Jonny Kong, Sizhe Wang, Yufei Feng, Phuc Dinh, Y. Charlie Hu, and Dimitrios Koutsonikolas, *Proc. of ACM IMC*, 2025.
- 2025 Root Cause Analysis of Cellular Network Throughput Degradations under Vehicular Mobility,
Eduardo Baena, Zihan Xu, Moinak Ghoshal, **Imran Khan**, Phuc Dinh, Z. Jonny Kong, Y. Charlie Hu, and Dimitrios Koutsonikolas, *Proc. of IEEE MASS*, 2025.
- 2024 An open, programmable, multi-vendor 5g o-ran testbed with nvidia arc and openairinterface,
Davide Villa*, **Imran Khan***, Florian Kaltenberger, Nicholas Hedberg, Rúben Soares Da Silva, Anupa Kelkar, Chris Dick, Stefano Basagni, Josep M. Jornet, Tommaso Melodia, Michele Polese, and Dimitrios Koutsonikolas, *IEEE INFOCOM 2024 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2024.
- 2024 An Experimental Study of Low-Latency Video Streaming over 5G,
Imran Khan, Tuyen X Tran, Matti Hiltunen, Theodore Karagioules, and Dimitrios Koutsonikolas, *Proc. of IEEE MeditCom*, 2024.
- 2024 How Mature is 5G Deployment? A Cross-Sectional, Year-Long Study of 5G Uplink Performance,
Imran Khan, Moinak Ghoshal, Joana Angjo, Sigrid Dimce, Mushahid Hussain, Paniz Parastar, Yenchia Yu, Claudio Fiandrino, Charalampos Orfanidis, Shivang Aggarwal, Ana C. Aguiar, Ozgu Alay, Carla F. Chiasserini, Falko Dressler, Y. Charlie Hu, Steven Y. Ko, Dimitrios Koutsonikolas, and Joerg Widmer, *Proceedings of the IFIP/IEEE Networking 2024 Conference*, 2024.
- 2024 On the Predictability of Fine-grained Cellular Network Throughput using Machine Learning Models,
Imran Khan*, Omar Basit*, Phuc Dinh*, Z. Jonny Kong*, Y. Charlie Hu, Dimitrios Koutsonikolas, Myungjin Lee, and Chaoyue Liu, *Proc. of IEEE MASS*, 2024.
- 2023 Performance of Cellular Networks on the Wheels,
Imran Khan*, Moinak Ghoshal*, Z. Jonny Kong*, Phuc Dinh*, Jiayi Meng, Y. Charlie Hu, and Dimitrios Koutsonikolas, *Proc. of ACM IMC*, 2023.
- 2023 Can 5G mmWave Enable Edge-Assisted Real-Time Object Detection for Augmented Reality?,
Moinak Ghoshal, Z. Jonny Kong, Qiang Xu, Zixiao Lu, Shivang Aggarwal, **Imran Khan**, Jiayi Meng, Yuanjie Li, Y. Charlie Hu, and Dimitrios Koutsonikolas, *2023 31st International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS)*, 2023.
- 2022 An In-Depth Study of Uplink Performance of 5G mmWave Networks,
Moinak Ghoshal, Z. Jonny Kong, Qiang Xu, Zixiao Lu, Shivang Aggarwal, **Imran Khan**, Yuanjie Li, Y. Charlie Hu, and Dimitrios Koutsonikolas, *Proc. of the ACM SIGCOMM 5G-MeMU Workshop*, 2022.
- 2021 Multipath TCP in Smartphones Equipped with Millimeter Wave Radios,
Imran Khan, Moinak Ghoshal, Shivang Aggarwal, Joerg Widmer, and Dimitrios Koutsonikolas, *Proc. of ACM WINTech '21*, 2021.

Journal Articles

- 2025 X5G: An Open, Programmable, Multi-Vendor, End-to-End, Private 5G O-RAN Testbed With NVIDIA ARC and OpenAirInterface, Davide Villa*, **Imran Khan***, Florian Kaltenberger, Nicholas Hedberg, Rúben Soares da Silva, Stefano Maxenti, Leonardo Bonati, Anupa Kelkar, Chris Dick, Eduardo Baena, Josep M. Jornet, Tommaso Melodia, Michele Polese, and Dimitrios Koutsonikolas, *IEEE Transactions on Mobile Computing*, 2025.
- 2025 How mature is 5G deployment? A cross-sectional, year-long study of 5G uplink performance, **Imran Khan**, Moinak Ghoshal, Joana Angjo, Sigrid Dimce, Mushahid Hussain, Paniz Parastar, Yenchia Yu, Xueting Deng, Sumit Hawal, Shirui Huang, Ameya Rane, Yin Wang, Claudio Fiandrino, Charalampos Orfanidis, Shivang Aggarwal, Ana C. Aguiar, Ozgu Alay, Carla Fabiana Chiasserini, Falko Dressler, Y. Charlie Hu, Steven Y. Ko, Dimitrios Koutsonikolas, and Joerg Widmer, *Computer Communications*, 2025.
- 2022 Musher: An agile multipath-tcp scheduler for dual-band 802.11ad/ac wireless lans, Shivang Aggarwal, Swetank Kumar Saha, **Imran Khan**, Rohan Pathak, Dimitrios Koutsonikolas, and Joerg Widmer, *IEEE/ACM Transactions on Networking*, 2022.

Posters/Demos

- 2024 [Demo] An Open, Programmable, Multi-Vendor 5G O-RAN Testbed with NVIDIA ARC and OpenAirInterface, Davide Villa, **Imran Khan**, Florian Kaltenberger, Nicholas Hedberg, Ruben Soares Da Silva, Anupa Kelkar, Chris Dick, Stefano Basagni, Josep M. Jornet, Tommaso Melodia, Michele Polese, and Dimitrios Koutsonikolas, *ACM WiNTECH*, 2024.
- 2022 [Demo] NextG-UP: A longitudinal and cross-sectional study of uplink performance of 5G networks, Moinak Ghoshal, **Imran Khan**, Qiang Xu, Z. Jonny Kong, Y. Charlie Hu, and Dimitrios Koutsonikolas, *ACM MobiCom*, 2022.

Skills

- Languages Python, C++, Java/Android, Bash, SQL, HTML, CSS
- Tools XCAL-Accuver (5G/LTE signaling message analysing tool) || Keysight RuSIM || Spectrum Analyzer || Git || Wireshark || Scikit-Learn || PyTorch || Keras || Video Streaming tools (Puffer, FFmpeg)
- Platform Linux || Android || Docker || Cloud (AWS, GCP)
- Protocols NR/LTE 3GPP standards || TCP/MPTCP/UDP protocols and their implementation (Linux Source Codes) || /Standards IEEE 802.11 ax/ac/ad/b/g/n standards

Source Codes & Datasets

- 2024 **Dataset** for Cross-Sectional, Year-Long Study of 5G Uplink Performance (IFIP).
- 2024 **Dataset** for Predictability of Fine-Grained Cellular Network Throughput Using ML Models (IEEE MASS).
- 2023 **Dataset** for 5G Network Performance on the Wheels (ACM IMC).
- 2024 **NextG-UP** Android App for Crowdsourced 5G Uplink Performance Measurements.
- 2021 MPTCP **Source Code** for Google Pixel 5.
- 2021 MPTCP **Source Code** for ASUS ROG Phone 2 (WiNTECH).

Professional Services/Awards

- 2026 COMCOM, Computer Networks, IEEE Transactions on Green Comm. and Networking, *Reviewer*.
- 2025 COMCOM, IEEE VTC, IEEE Transactions on Green Comm. and Networking, *Reviewer*.
- 2024 Computer Networks, IEEE IoT, IEEE ICC, IEEE WCNC, *Reviewer*.
- 2024 IEEE LANMAN, IEEE WOWMOM, *Web Chair*.
- 2023 IEEE WCNC, *TPC Member*.
- 2021 NSF Funded Student travel grant, *ACM Mobicom*.
- 2019 NSF Funded travel grant , *POWDER Network and Wireless Week*.